

# The role of fecal microbiota transplantation in the treatment of acute graft-versus-host disease

## ABSTRACT

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is one of the most important methods for treating a wide range of hematologic malignancies and bone marrow failure diseases. However, graft-versus-host disease (GVHD), a major complication associated with this method, can seriously affect the survival and quality of life of patients. Acute GVHD (aGVHD) occurs within 100 days after transplantation, and gastrointestinal aGVHD (GI-aGVHD) is one of the leading causes of nonrecurrent death after allo-HSCT. In recent years, fecal microbiota transplantation (FMT) has been attempted as an emerging treatment method for various diseases, including aGVHD after HSCT. Studies have shown encouraging preliminary clinical results after the application of FMT in aGVHD, particularly steroid-resistant aGVHD. Additionally, several studies have demonstrated that the gut microbiota plays an important immunomodulatory role in the pathogenesis of GVHD. Consensus guidelines recommend FMT as a secondary option for the treatment of aGVHD. This article aims to review FMT treatment for GI-aGVHD after allo-HSCT.

**KEY WORDS:** aGVHD, allo-HSCT, FMT, intestinal microecology

## INTRODUCTION

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is one of the most important methods to treat a wide range of hematologic malignancies and bone marrow failure diseases.<sup>[1-3]</sup> Graft-versus-host disease (GVHD), a major complication associated with this method, can seriously affect the survival and quality of life of patients. Acute GVHD (aGVHD) occurs within 100 days after transplantation, and gastrointestinal aGVHD (GI-aGVHD) is one of the leading causes of nonrecurrent death after allo-HSCT.<sup>[4,5]</sup> Therefore, the treatment of aGVHD, especially GI-aGVHD, is vital.

In recent years, fecal microbiota transplantation (FMT) has been attempted as an emerging treatment method for various diseases, including

aGVHD after HSCT. Previously, we suggested FMT as a secondary option for the treatment of aGVHD in both consensus guidelines.<sup>[6,7]</sup> Studies have shown encouraging preliminary clinical results after applying FMT in aGVHD, especially steroid-resistant aGVHD. Furthermore, several studies have shown that the gut microbiota plays an important immunomodulatory role in the pathogenesis of GVHD. FMT is an effective method to re-establish the intestinal microecology.<sup>[8]</sup> It involves making suitable preparations of the feces of healthy people and delivering them to repair or rebuild the patient's intestinal flora and treat pathogenic infections, metabolic disorders, and other diseases. FMT is considered a good source of probiotics for restoring intestinal microecological diversity, and its efficacy in treating refractory or recurrent *Clostridium difficile* infections is well established.<sup>[9]</sup> This method has been applied for the treatment of digestive diseases (such as chronic hepatitis B and inflammatory bowel disease),<sup>[10]</sup> neuropsychiatric

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Zhi Guo<sup>1,2\*</sup>,  
Mingxin He<sup>2\*</sup>,  
Liang Shao<sup>3\*</sup>,  
Yue Li<sup>2</sup>,  
Xiaochen Xiang<sup>1</sup>,  
Qiang Wang<sup>1</sup>

<sup>1</sup>Institute of Infection, Immunology and Tumor Microenvironment, Hubei Province Key Laboratory of Occupational Hazard Identification and Control, School of Medicine, Wuhan University of Science and Technology, Wuhan, <sup>2</sup>Department of Hematology, Huazhong University of Science and Technology Union Shenzhen Hospital, Shenzhen, <sup>3</sup>Department of Hematology, Zhongnan Hospital of Wuhan University, Wuhan, China  
\*Co-first author.

**For correspondence:** Dr. Zhi Guo, Institute of Infection, Immunology and Tumor Microenvironment, Hubei Province Key Laboratory of Occupational Hazard Identification and Control, School of Medicine, Wuhan University of Science and Technology, Wuhan - 430065, China; Department of Hematology, Huazhong University of Science and Technology Union Shenzhen Hospital, Shenzhen - 518052, China.  
E-mail: guozhi77@126.com

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Prof. Qiang Wang,

Institute of Infection, Immunology and Tumor Microenvironment, Hubei Province Key Laboratory of Occupational Hazard Identification and Control, School of Medicine, Wuhan University of Science and Technology, Wuhan - 430065, China.

E-mail: wangqiang@wust.edu.cn

diseases (such as Alzheimer's disease),<sup>[11]</sup> hematologic diseases (such as aGVHD),<sup>[12]</sup> metabolic syndromes (such as glycosuria),<sup>[13]</sup> and multidrug-resistant bacteria (such as broad-spectrum  $\beta$ -lactamase-producing *Escherichia coli*, carbapenemase-producing *Pseudomonas aeruginosa*, and vancomycin-resistant *Enterococcus faecalis*).<sup>[12]</sup> This article aims to review FMT treatment for GI-aGVHD after allo-HSCT.

### aGVHD and the intestinal microecology

#### aGVHD

The incidence of aGVHD in patients with allo-HSCT is approximately 50%, and that of moderate-to-severe aGVHD ranges from 13% to 47%; the main clinical manifestations of aGVHD include diarrhea and skin rashes on the limbs and trunk, with abdominal pain and intestinal obstruction in severe cases.<sup>[4,14]</sup> aGVHD mainly involves the skin, GI tract, and liver.<sup>[15]</sup> The disease mainly occurs due to the recognition of T-cells in the grafts by the donor's own or the recipient's antigen-presenting cells, followed by their activation, differentiation, and migration to the recipient's target organs, including the GI tract, lungs, liver, and skin, causing organ damage or failure. Transplantation preconditioning is the initiator of aGVHD. Radiotherapy or chemotherapy causes inflammatory injury in recipient tissues (e.g., the gastrointestinal mucosa, skin, and liver), leading to the release of large amounts of inflammatory factors (such as IL-1, TNF- $\alpha$ , and IL-6), PAMPs, and damage-associated molecular patterns and an increase in the expression levels of major histocompatibility complex antigens and costimulatory molecules; this results in the activation of the donor T-cells in the grafts, which differentiate, proliferate, and attack the recipient target organ.<sup>[16-18]</sup> According to the Chinese expert consensus on allogeneic hematopoietic stem cell transplantation for the treatment of hematological diseases III - Acute GVHD (2020 version), the first-line treatment for aGVHD is glucocorticoids, commonly used with methylprednisolone. The second-line treatment includes an IL-2 receptor monoclonal antibody (baliximab), MTX, or lucitinib, and the third-line treatment includes ATG, MSC, and FMT, among others.

#### Intestinal microecology overview

Microecology refers to the totality of microorganisms in a specific environment or ecosystem, including their cell populations, quantity, and genetic materials (genomes), which defines the microbiota and all their genetic and physiological functions; their connotations include the interactions between the microorganisms and their environments and hosts.<sup>[19,20]</sup> There are more than 100 trillion symbiotic microorganisms on the surface and in a healthy human body, mainly distributed in the GI tract, oral cavity, skin, respiratory tract, urogenital

tract, and other parts.<sup>[21]</sup> The microbial flora of healthy people, especially the intestinal flora, is considered an "essential organ," and its genome is about 150 times larger than that of the whole human genome.<sup>[22]</sup> Under physiological conditions, intestinal microorganisms are a highly diverse group of nonpathogenic symbiotic bacteria, mainly consisting of Bacteroidetes and the Firmicutes, Actinobacteria, and Proteobacteria phyla.<sup>[23]</sup> Among them, 1000 microbial species reside within the small intestine, with the Firmicutes and Bacteroidetes phylums predominating and accounting for about 90% of the total.<sup>[24]</sup>

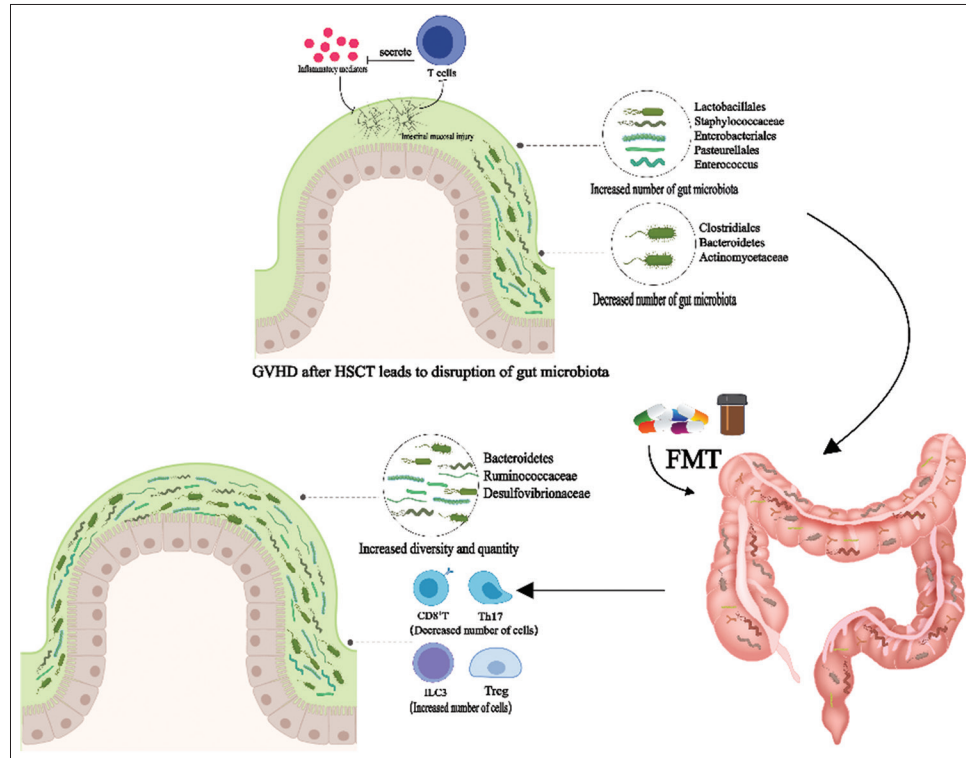
#### aGVHD and intestinal microecology overview

Patients receiving allo-HSCT are prone to dysbiosis due to the destruction of the intestinal tract after pretransplant radiotherapy or chemotherapy.<sup>[25,26]</sup> Loss of intestinal microbial diversity and alterations in the microbial composition occur after allo-HSCT, particularly in patients who develop aGVHD.<sup>[27]</sup> Taur *et al.*<sup>[28]</sup> divided the population into three groups of low, intermediate, and high diversity based on the content of the intestinal flora in post-transplant recipients. The 3-year overall survival (OS) of patients in the three groups was 36%, 60%, and 67% ( $P = 0.019$ ), and the 3-year treatment-related mortality was 53%, 23%, and 9%, respectively. A study comprising 1362 patients with allo-HSCT showed that the higher the degree of diversity in the patient's flora, the lower the patient's GVHD-related mortality rate; the lower the degree of diversity, the higher the mortality rate.<sup>[29]</sup> An imbalance of the intestinal flora during allo-HSCT can be manifested as a change in the composition of the flora. Among them, *Lactobacillales*, *Staphylococcaceae*, *Enterobacteriales*, and *Pasteurellales* are most significantly amplified in aGVHD patients.<sup>[30]</sup> Studies have shown that the amplification of *Enterococcus* increases the severity and mortality of aGVHD; the role of *Lactobacillus* remains controversial.<sup>[30]</sup> Taur *et al.* reported that the enrichment of *Enterobacteriales* was positively associated with GVHD-related mortality.<sup>[28]</sup> *Pasteurella* is seen in pediatric patients who develop GI-aGVHD and is strongly associated with the severity of diarrhea.<sup>[31]</sup> In contrast, *Clostridiales*, *Bacteroidetes*, and *Actinomycetaceae* are significantly reduced in aGVHD patients.<sup>[32,33]</sup> Sofi *et al.*<sup>[34]</sup> found that an increase in Bacteroidetes in the small intestine was closely related to an improvement in GVHD; additionally, treatment with *Bacteroides fragilis* increased the diversity of the intestinal flora, augmented the short-chain fatty acids (SCFA), IL-22 and Tregs, and attenuated GVHD. Table 1 summarizes the relationship between aGVHD and the gut microbiota after allo-HSCT. Figure 1 shows the role of FMT in GVHD after HSCT.

### The role of gut microbiota in aGVHD

The intestinal flora plays an important role in the pathophysiologic process of GVHD. HSCT affects the body's intestinal microecology, which may also influence the prognosis of HSCT through multiple mechanisms. During

transplantation, the composition of the original intestinal flora is altered, and the number of intestinal symbiotic bacteria decreases due to the high intensity of the pretreatment chemotherapy and the use of preventive antibiotics, resulting in a microecological imbalance or bacterial translocation;



**Figure 1:** The role of FMT in GVHD after HSCT. The process of GVHD can be characterized by impaired intestinal mucosal barrier function, abnormal immune regulation, activation of T-lymphocytes, release of inflammatory mediators, damage to target organs, and imbalance of the original gut microbiota. FMT controls GVHD by restoring the diversity of gut microbiota and immune regulation

**Table 1: Association between aGVHD and gut microbiota after allo-HSCT**

| Patients                                   | Results                                                                                                                                                                                                                                                                                                                                                  | Sample types and detection techniques | References |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------|
| 1362 patients undergoing allo-HSCT         | The risk of GVHD-related death was lower among patients with higher intestinal diversity than among those with lower diversity.                                                                                                                                                                                                                          | Stool, 16S rRNA gene sequencing       | [29]       |
| 10 pediatric patients undergoing allo-HSCT | The onset of aGVHD is associated with specific gut microbiota signatures. Non-aGVHD patients showed higher abundances of propionate-producing Bacteroidetes.                                                                                                                                                                                             | Stool, 16S rRNA gene sequencing       | [33]       |
| 36 pediatric patients undergoing allo-HSCT | This aGVHD-predisposing ecosystem state was characterized by (i) reduced diversity, (ii) lower <i>Blautia</i> content, and (iii) an increase in <i>Fusobacterium</i> abundance.                                                                                                                                                                          | Stool, 16S rRNA gene sequencing       | [35]       |
| 141 patients undergoing allo-HSCT.         | The diversity was lower in the aGVHD group than the non-aGVHD group, and the <i>Lachnospiraceae</i> family of <i>Clostridiales</i> was decreased in the aGVHD group. The abundances of <i>Blautia</i> , <i>Lachnoclostridium</i> , <i>Erysipelatoclostridium</i> , and <i>Eubacterium</i> were lower in the aGVHD group than the non-aGVHD group.        | Stool, 16S rRNA gene sequencing       | [36]       |
| 81 patients undergoing allo-HSCT           | The aGVHD group had a greater low microbial diversity than the non-aGVHD group. The bacterial community was depleted of Clostridia (e.g., the <i>Lachnospiraceae</i> and <i>Ruminococcaceae</i> families) and enriched for <i>Gammaproteobacteria</i> (e.g., the <i>Enterobacteriaceae</i> family) in the aGVHD group compared with the non-aGVHD group. | Stool, 16S rRNA gene sequencing       | [37]       |
| 31 patients undergoing allo-HSCT           | Dysbiosis of the intestines and acute GVHD occurred simultaneously, and <i>Enterococcus</i> species were abundant.                                                                                                                                                                                                                                       | Stool, 16S rRNA gene sequencing       | [38]       |

consequently, the diversity and stability of the intestinal flora are destroyed, resulting in an imbalance or bacterial translocation of the original intestinal flora.<sup>[39]</sup> Figure 2 demonstrates the effect of antibiotics on human gut microbiota and the form of regulation. Bacteria that dominate the intestinal flora include short-chain fatty acid-producing strains specialized in the fermentation of oligosaccharides, such as bifidobacteria.<sup>[40,41]</sup> The colonization or translocation of various microbiota in the intestinal tract often precedes the onset of an infection and is one of the common causes of bacteremia and septicemia. The intestinal flora plays a crucial role in intestinal homeostasis and immune regulation and has been recognized as a predictor of clinical outcomes in patients receiving allo-HSCT.<sup>[42]</sup> A change in the intestinal flora may determine the severity of GVHD. After transplantation, the composition of the intestinal flora changes, resulting in flora imbalance. The intestinal microorganisms easily penetrate the damaged intestinal mucosa, causing abnormal immune responses, activating T-lymphocytes, promoting the release of inflammatory mediators, and damaging the GI mucosal barrier, thereby damaging target organs, such as the GI tract.<sup>[7,23]</sup>

### Application of intestinal microbial detection technology in aGVHD

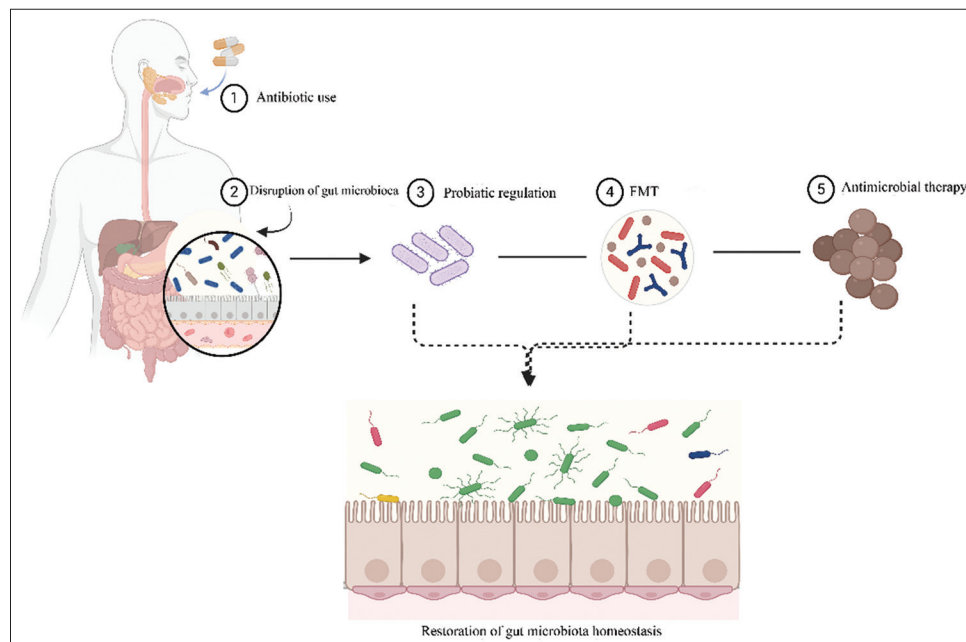
#### Technology used to analyze the gut microbial profile

The commonly used techniques for microbial spectrum analysis are 16S rRNA and metagenomic sequencing.<sup>[43,44]</sup> *In situ* hybridization and polymerase chain reaction (PCR) have been used to identify and quantify bacteria. The gene encoding the 16S ribosomal RNA subunit contains regions that are highly conserved among different bacterial species and nine highly variable regions unique to specific bacterial species, and bacterial communities can be identified by sequencing

these variable regions.<sup>[45,46]</sup> This method can be used to identify bacterial taxa, but it does not definitively identify the bacterial strain. The classification and identification of bacterial communities require reference to databases, and open-source software such as Mothur and QIIME can be used to analyze 16S rRNA sequencing data.<sup>[47]</sup> Metagenome sequencing can provide unbiased sequencing of all genetic material in a sample, and extracted gDNA is massively sequenced in parallel to provide broad coverage of the microbial community.<sup>[48-50]</sup> Metagenome sequencing can detect microorganisms with extremely low abundance due to high sequence coverage.<sup>[51]</sup> Microbial metagenome sequencing and metabolomics profiling are important tools for understanding the correlation between taxon level changes and metabolic functions in the host.<sup>[52]</sup>

#### Diagnosis of microbial community changes after GVHD using biomarkers

The microbiota is altered in the presence of GVHD or disease recurrence after allo-HSCT; hence, detecting changes in the abundance of one or more bacteria (e.g. *Blautia* and *E.limosum*) may prove helpful.<sup>[53]</sup> In addition, several clinical reports have suggested that the genotyping of fucosyltransferase 2 (FUT2) can serve as a biomarker for the development of GVHD and bacteremia after allo-HSCT.<sup>[54]</sup> The FUT2 genotype may affect the risk of Crohn's disease through changes in the glycosylation of intestinal surface proteins and the host microbiota.<sup>[55]</sup> In addition, 3-3-indophenol sulfate, a urinary metabolite produced by *Clostridium*, is another potential biomarker for the health of the intestinal flora.<sup>[56]</sup> It is a prognostic indicator for the decrease in the incidence of intestinal GVHD and improvement in the patient survival rate; moreover, it is related to the reduction in intestinal inflammation.<sup>[56]</sup> Considering the rapidity of the 3-indolyl



**Figure 2:** Effects of antibiotics on human gut microbiota and forms of regulation



sulfate level detection test, it may be used to assess the health of a patient's microbiota, with the clinical significance of evaluating the recovery of damaged microbiota during GVHD.<sup>[57]</sup>

#### *Other detection techniques*

Changes in the intestinal flora can be used as a noninvasive biomarker of GVHD. In addition to 16S rRNA and metagenome sequencing techniques, quantitative PCR (qPCR) can provide valuable information about the relative abundance and dynamics of bacterial communities.<sup>[58]</sup> In clinical patients, especially in high-risk patients, qPCR can be used for the real-time monitoring of specific microbial strains before, during, and after allo-HSCT.<sup>[59]</sup> Furthermore, evaluating the damage to the intestinal microflora may support using narrow-spectrum antibiotics instead of broad-spectrum antibiotics because broad-spectrum antibiotics will cause more collateral damage to the intestinal mucosa and promote the emergence of drug-resistant microbial strains.<sup>[60-63]</sup> Therefore, targeted therapy, such as coupling pathogen-specific antibodies with antibiotics, is necessary for certain gut microbiota. Specific pathogens can be targeted using CRISPR-Cas9 phage. In terms of anticolonization, genetically engineered bacteria can be used to compete with pathogens, and ligands or pharmacological compounds for intestinal antimicrobial peptides and Toll-like receptors can help reduce the colonization of intestinal pathogenic microorganisms, reducing the risk of GVHD. High-throughput culture techniques, such as matrix-assisted laser-resolved ionization time-of-flight mass spectrometry, have been used to determine the microbial composition; however, nonculturable microorganisms are missed by this technique. Other *in vitro* systems, such as the coculture of stem cell-derived intestinal epithelial cells with the microbiome, may be useful for studying the interactions between microbial cultures and human intestinal cells.<sup>[64]</sup>

#### *Gut microbiota and immune reconstitution and follow-up*

The gut microbiota may affect immune reconstitution after allo-HSCT and serve as a monitoring indicator to adjust immune-related strategies after transplantation.<sup>[65,66]</sup> Defective hematopoiesis has been reported in germ-free mice, suggesting that gut microbiota may promote the development of homeostatic myeloid cells and increase the potential for differentiation to granulocytes.<sup>[67]</sup> How the gut microbiota controls immune responses through distant sites, such as the bone marrow, remains poorly understood. A study involving a syngeneic mouse transplantation model reported the occurrence of delayed immune reconstitution in antibiotic-treated transplant recipients, thus indicating that changes in the microbiota impact hematopoiesis and immune reconstitution in allo-HSCT recipients. The gut microbiota can modulate host immunity, can play a role in the development of GVHD, and is associated with the severity of GVHD; therefore, indicators related to alterations in the gut microbiota may prove useful.<sup>[68]</sup> An increase in the gut microbiota load, such as an abundance of *E. coli*, may indicate an increased risk

of bacteremia. Surveillance of intestinal Enterobacteriaceae strains (e.g., *E. coli*, *Klebsiella* spp., and Enterobacteriaceae) in allo-HSCT patients may help in identifying patients at risk for infections and reduce the risk of Enterobacteriaceae bacteremia.<sup>[69]</sup> The interaction between microbiota and host immunity can have long-term effects and be used to identify patients at risk of disease recurrence.<sup>[70]</sup>

#### **Clinical application of FMT in the treatment of aGVHD**

In 2016, Japanese scholars Kakihana *et al.*<sup>[71]</sup> applied FMT for the first time to treat glucocorticoid-resistant/drug-resistant aGVHD in four patients. All four patients responded to the treatment, of which three achieved complete remission (CR), and one had partial remission; the hormone-resistant patients showed improvement in the GI symptoms within a few days, and no serious toxic side effects were observed during the treatment.<sup>[71]</sup> Those who responded to FMT treatment had a significant increase in the number of Treg cells. Although the number of patients included in the study was limited, it was a precedent for the treatment of FMT in GI-aGVHD. Subsequently, Australian scholars Spindelboeck *et al.*<sup>[72]</sup> studied nine patients with refractory severe GI-aGVHD and found that four responded to treatment with FMT and achieved a significant survival benefit. The study also found that concomitant use of antibiotics was a major factor for the failure of FMT treatment. The use of feces from healthy individuals significantly hampered the establishment of microbial colonization profiles in FMT-treated individuals. The *in vivo* microbial colony profiles of those who responded to FMT treatment were almost the same as those of the feces provider.<sup>[72]</sup> Moreover, patients who responded to FMT treatment had a significant decrease in the number of CD8<sup>+</sup> T-cells and Th17 cells to near normal levels and a significant increase in the number of Tregs and ILC3 cells; the trend was reversed in those who did not respond to FMT treatment. This study suggests that FMT repairs the diversity of the gut microbiota and alters the immune status of GI-aGVHD patients.

Chinese scholars Qi *et al.* treated eight patients with hormone resistant GI aGVHD using China fmtBank.<sup>[73]</sup> All eight patients had relief from the clinical symptoms after receiving the first FMT. Two patients had complete cessation of diarrhea, and five patients had complete relief from abdominal pain after 2 weeks. The diversity of the flora in the patients before FMT treatment was reduced; additionally, there was an imbalance in the proportion of Firmicutes and Proteobacteria and a reduction in the proportion of Bacteroidetes. However, 1 week after FMT treatment, the patients presented with high levels of *Bacteroidetes*, *Ruminococcaceae*, and *Desulfovibrionaceae*. Beneficial species such as *Bacteroidetes* spp. were predominant after FMT. The 90-day progression-free-survival analysis showed that the FMT group was significantly better than the group that did not receive FMT treatment. However, there was no significant difference in OS between the two groups. No serious FMT-related side effects were found throughout the treatment.

Qi *et al.* initiated a nonrandomized, open, I/II clinical study enrolling 41 investigators using FMT for the treatment of hormone-resistant GI-GVHD.<sup>[73]</sup> Twenty-three hormone-resistant patients with IV° GI-GVHD were treated with FMT, while the remaining 18 served as the control group. The results of the study showed that the clinical symptom remission rate was significantly higher in the FMT-treated group than in the control group on the 14<sup>th</sup> and 21<sup>st</sup> days of FMT treatment. After 90 days of follow-up, the FMT-treated group had a better OS than the control group. The mean survival time at the end of the study was >539 and 107 days for the two groups, respectively, and the FMT-treated group had significantly better event-free survival (EFS) and OS than the control group. No side effects significantly associated with FMT were found in this study. Shouval *et al.*<sup>[74]</sup> reported for the first time the application of FMT using the oral capsule route in seven patients with hormone-resistant/dependent GI-aGVHD. Two patients achieved CR, and the diversity of the gut microbiota was improved, thus indicating that the application of FMT via an oral capsule is safe and effective for treating GI-aGVHD.

The results of the above-mentioned studies indicate the safety of FMT for the treatment of GI-aGVHD; it improves the diversity of the patient's gut microbiota and changes its composition to achieve the therapeutic effects. However, can FMT be used in combination with other treatments? Liu *et al.*<sup>[75]</sup> tried a combination of FMT and ruxolitinib as a salvage treatment in 21 allo-HSCT-induced hormone-resistant III–IV° GI-aGVHD patients. The objective response rate (ORR) on the 28<sup>th</sup> day of treatment was 71.4%, with ten patients achieving CR. On the 56<sup>th</sup> day, the duration of response was 80%, and the recurrence rate of GVHD in responsive patients was 33.3%. The study found that the microbial diversity of the patients had significantly improved after FMT treatment. The main side effects of combination therapy were viral reactivation (61.9%) and severe hemocytopenia (81%). The predicted 6-month OS and EFS were 57.1% and 52.4%, respectively. These data

suggest that FMT can be used in combination with ruxolitinib for the clinical treatment of hormone-resistant GI-aGVHD. Table 2 summarizes the clinical trials of FMT intervention for the treatment and prevention of aGVHD caused by allo-HSCT.

### The treatment process and methods of FMT

#### Screening of donors

According to the European consensus report on the clinical application and operation of fecal bacteria transplantation, all the following criteria were required during the selection of donors<sup>[79]</sup>:

- (i) Drug use: No antibiotic treatment, immunosuppressant treatment, or chemotherapy in the last 3 months and not treated with proton-pump inhibitors recently.
- (ii) Infectious diseases: No history or exposure history of HIV, HBV, HCV, HEV, and tuberculosis; no systemic infections that have not yet been fully controlled; no use of prohibited drugs such as drugs; no history of tissue or organ transplant; no blood transfusion within 1 year; no recent infections of parasites, norovirus, and flagellates; and no history of vaccination with live attenuated vaccines over the past 6 months.
- (iii) Gastrointestinal, endocrine, and neurological diseases: No inflammatory bowel disease, irritable bowel syndrome, chronic functional constipation, and other chronic intestinal dysfunction; no chronic systemic autoimmune disease with intestinal involvement; no history of gastrointestinal tumors and polyps; no recent diarrhea or blood in stool; no history of neurological disease or neurodegenerative lesions; no mental illness; and a body mass index of  $\leq 25$ .

#### Serological and fecal screening programs for donors

It is recommended that the routine serologic tests include (i) screening for various common viruses such as CMV, EBV, HAV, HBV, HCV, HEV, HIV, and syphilis; (ii) routine blood tests, CRP + ESR, liver, and kidney glucose

**Table 2: Clinical trial of FMT intervention for the treatment and prevention of aGVHD caused by all-HSCT**

| Patients                                                                           | Results                                                                                                                                                                                                                                  | FMT input pathway | References |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|
| Four patients with GI-aGVHD (steroid-resistant, $n=3$ ; steroid-dependent, $n=1$ ) | FMT was safely performed in patients, with three complete and one partial responses.                                                                                                                                                     | Nasoduodenal tube | [71]       |
| 10 patients with treatment-refractory GI-GVHD                                      | Four of the 10 patients treated with FMT had sustained remission of GI-GvHD.                                                                                                                                                             | Endoscope         | [72]       |
| Eight patients with GI-aGVHD                                                       | After receiving the FMTs, the clinical symptoms of all patients were relieved, bacteria were enriched, and microbiota compositions were reconstructed.                                                                                   | Nasoduodenal tube | [73]       |
| 41 patients with steroid-refractory GI-GVHD (FMT, $n=23$ ; placebo, $n=18$ )       | Clinical remission was significantly greater in the FMT group than in the control group. The FMT group showed a better overall survival (OS). The median survival time was >539 days in the FMT group and 107 days in the control group. | Unspecified       | [68]       |
| 26 patients with aGVHD (FMT, $n=18$ ; placebo, $n=8$ )                             | FMT may have protective effects against aGVHD, especially in patients with more severe microbiota disruptions.                                                                                                                           | Oral capsule      | [76]       |
| Two patients with aGVHD                                                            | In the first patient, FMT resulted in the complete resolution of symptoms. FMT eliminated drug-resistant pathogens, namely very drug-resistant <i>Enterococcus</i> spp.                                                                  | Intranasal probe  | [77]       |
| 11 patients with aGvHD (repeated in 3 patients)                                    | The overall response rate (ORR) in cases of aGVHD reached 57%, including complete response (CR) in 42% of procedures.                                                                                                                    | Nasoduodenal tube | [78]       |

and electrolytes. It is recommended that routine fecal examination includes (i) screening for various pathogenic bacteria, including *Clostridium difficile*, *Salmonella*, *Shigella*, *Campylobacter*, O157 H7 *E. coli*, *Yersinia*, vancomycin-resistant enterococci, methicillin-resistant *Staphylococcus aureus*, and Gram-negative multidrug-resistant bacteria; (ii) viral screening, e.g., norovirus; (iii) parasitic screening, such as *Giardia*, *Cryptosporidium*, *Blastocystis hominis*, and helminths; and (iv) fecal occult blood test.

#### *Key points for preparing fecal microbiota*

The key points in the preparation of fresh fecal material include the following: (i) Fresh feces must be used within 6 hours of defecation, (ii) the preparation time should be as short as possible to ensure bacterial integrity, (iii) anaerobic storage and processing should be used wherever possible, (iv) at least 30 g of feces is required, (v) the feces should be resuspended in saline using a mixer/stirring rod and after mixing the feces should be passed through a strainer to remove residues and small particles to avoid clogging of the syringe and tubing during input, (vi) protective gloves and face screens should be worn during preparation, and (vii) the preparation site should be strictly disinfected before and after preparation to avoid spillage.

The key points in the preparation of frozen fecal material include the following: (i) At least 30 g of fresh feces should be added to 150 ml of normal saline; (ii) before freezing, glycerol should be added to make the final concentration glycerol of 10%; (iii) the final fecal suspension obtained should be labeled in detail and frozen at  $-80^{\circ}\text{C}$ ; (iv) on the day of reinfusion, the fecal bacteria storage solution should be placed in a water bath at  $37^{\circ}\text{C}$  to thaw and then infused within 6 hours after thawing; (v) normal saline may be added appropriately to achieve the expected concentration after thawing; and (vi) repeated freezing and thawing must be avoided.

#### *Input pathways and security*

The input pathways in the FMT treatment process are as follows:

- (i) Upper gastrointestinal pathway: Mainly through gastroscopically reinfusion. The disadvantage is that the amount of fecal bacterial suspension input is relatively small, and a nasojejunal tube must be placed to reinfuse 500 ml.
- (ii) Colonoscopy pathway: Fecal bacterial suspension is injected through a colonoscope.
- (iii) Enema pathway: Suitable for frail patients who cannot undergo colonoscopy.
- (iv) Oral capsules.

The short-term adverse reactions of FMT include those related to the input pathway of FMT and the basic condition of the recipient, such as fever, diarrhea, abdominal distension, abdominal spasm, constipation, and vomiting. The potential

long-term adverse reactions include obesity, peripheral neuritis, Sjögren's syndrome, ITP, and rheumatoid arthritis.

#### **Prospect**

FMT has shown initial efficacy in treating hormone-resistant GI-aGVHD after allo-HSCT. The main mechanisms of action may include the following: (i) restoring the diversity of the gut microbiota in patients, (ii) repairing the composition of the patient's gut microbiota, and (iii) altering the immune status of the body. Presently, clinical data from several countries, including ours, show that the adverse reactions of FMT are relatively small and basically controllable, indicating that its safety is acceptable. Despite a few randomized controlled clinical trials on the use of FMT in GVHD, there are more studies on the indications for FMT, and they are mainly focused on the effects of FMT on inflammatory bowel diseases (IBDs). A study demonstrated the safety and efficacy of FMT in treating recurrent or refractory *Clostridium difficile* infections (CDIs) in both IBD and non-IBD patients, and it also helped reduce IBD-associated inflammation. Recurrent CDI (rCDI), which is ineffective against conventional therapies, makes it the most common indication for FMT; however, exploration of other therapeutic uses for FMT is ongoing.<sup>[80-82]</sup> A first-of-its-kind study using FMT therapy in treating systemic lupus erythematosus (SLE) demonstrated that FMT restores the gut microbiota structure and reduces the active phase of lupus in SLE patients.<sup>[83]</sup>

FMT, as a new therapeutic approach, can influence the course of a wide range of chronic diseases, such as metabolic syndrome, malignancies, autoimmune diseases, and neurological disorders.<sup>[84]</sup> It is believed that FMT has a far-reaching application in treating GVHD in the near future. So far, the efficacy of FMT has been encouraging, with a few serious adverse events. If confirmed by randomized clinical trials, it is believed that FMT could become the standard treatment for steroid-refractory or dependent aGVHD. In the future, FMT may be more widely used in patients with aGVHD as a first-line treatment or preventive measure. In addition, every patient who receives an allo-HSCT will undergo gut microbiota analysis. FMT can be used to rebuild the gut microbiota and increase the integrity of the intestinal barrier, which is sufficient to prevent both acute and chronic GVHD. FMT can be used to reconstruct the gut microbiota and increase the integrity of the intestinal barrier, thereby preventing acute and chronic GVHD. Additional randomized clinical trials must be conducted to investigate and confirm these findings.

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### Conflicts of interest

There are no conflicts of interest.

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